

Perfect Subset Privacy for Data Sharing and Learning

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Abstract—As the size of modern datasets grows, it becomes increasingly common to delegate computational tasks to service providers. Doing so, however, raises privacy concerns. Privatization schemes which enable learning algorithms to be executed unaltered have been recently popularized under the name *instance encoding*, aiming to circumvent the large overhead of traditional cryptographic primitives. In this work we take an information/coding-theoretic approach towards instance encoding. Specifically, recent works have shown that general-purpose data sharing can be achieved without leaking information about any individual datapoint (marginally), while maintaining high mutual information with the dataset in its entirety. We first extend this framework to capture the entire privacy-utility tradeoff, accounting for the privatization of any subset of the dataset, and provide a coding scheme for doing so. Second, we introduce a necessary algebraic condition for applying unaltered learning algorithms on encrypted data, termed *signal preservation*, and present an additional scheme which guarantees it. Both schemes achieve almost maximal mutual information with the entire dataset, under appropriate assumptions. The construction relies on some classic ideas such as Shamir secret sharing, as well as a novel technique called random Hadamard coding.

I. INTRODUCTION

Collecting and processing large amounts of user data is central to many modern technologies. As data becomes more complex, there is a growing need for sharing it with a service provider that can cost-effectively store and process it. However, datasets often contain sensitive information, and as such, they must be privatized before being shared. In cases where the data is shared for learning, it is further desired to preserve an appropriate notion of learnability post privatization. The latter has been addressed in a series of recent works under the term *instance encoding* [1, 2, 4, 5, 9, 14] which refers to encrypting the dataset (into a privatized version thereof of the same size) prior to sharing it. By doing so, common learning algorithm can be executed *as-is*, as opposed to the large overhead induced by techniques such as homomorphic encryption [3, 8, 10] (see Section I-B for details).

In this paper, we address both the general-purpose and learning-oriented dataset privatization questions by proposing two schemes, one for each situation. Both schemes guarantee information-theoretic subset privacy, defined shortly, while maintaining almost maximal informativeness of the privatized dataset (up to a penalty due to data uniformization). The latter scheme also preserves an algebraic notion of learnability we call *signal preservation*.

A privatization scheme is a (possibly randomized) transformation of the original dataset $X^n := (X_1, \dots, X_n)$ into its privatized version $\tilde{X}^n$. A scheme is called *perfectly private* if the transformed and original datasets have zero mutual information, i.e., $I(X^n; \tilde{X}^n) = 0$. Clearly, perfect privacy is of no use for data sharing, since the data processing inequality (DPI) implies that privatized data cannot be used for inference on the original one. Yet, [11] recently showed that perfect privacy can be guaranteed for each *individual* datapoint, i.e., $I(X_i; \tilde{X}^n) = 0$, while maintaining high mutual information with the original dataset as a whole. This reveals a potential information-theoretic privacy-utility tradeoff. Can this instance be generalized to perfect privacy of any *subset* of datapoints up to a given size while maintaining high informativeness?

A. Contributions

We provide herein an affirmative answer to the question above, as delineated next. First, we propose a general-purpose data privatization mechanism that guarantees zero mutual information between the privatized data and any subset of up to r original datapoints. The mechanism first uniformizes the data by mapping it into a large enough finite field. The transformed datapoints are then linearly combined with random keys as in Reed-Solomon encoding so as to guarantee privacy, similar to Shamir secret sharing [12]. While this guarantees r -subset privacy, we simultaneously aim to keep the mutual information with the entire dataset, i.e., $I(X^n; \tilde{X}^n)$, as high as possible. We prove that the proposed schemes maximize this mutual information, subject to the subset privacy constraint and the uniformization mechanism. An attempt at furnishing such a scheme was made in [11, Corollary 4.1], but privacy guarantees were provided only to subsets of *consecutive datapoints*—a somewhat unnatural restriction. Our mechanism, which differs from that of [11], closes this gap, and enables privatization of arbitrary r -subsets.

We then move to the instance encoding setup, where the goal is to privatize the dataset such that it is not only informative but is also well adapted for learning. Motivated by the popularity of deep neural nets for statistical inference, we propose *signal preservation* as the desired notion of learnability. Here ‘signal’ refers to linear transformations of feature vectors, while ‘preservation’ means that any signal that can be obtained from the original data is also obtainable from the privatized version (with possibly different parameters). Signal

preservation guarantees that learning tasks that were possible on the original data are still possible after privatization.

Our second main contribution is an adaptation of the general-purpose scheme that ensures signal preservation in addition to the high mutual information with the entire dataset. The adaptation employs random Hadamard encoding in addition to the secret sharing techniques used for the general-purpose variant. To quantify the information loss, we set up a general information-theoretic framework for instance encoding and derive an upper bound on the mutual information between the clean and privatized datasets for any scheme in that framework. We then show that our proposed scheme achieves this upper bound up to a small penalty (that has to do with a data uniformization step). Altogether this produces an instance-encoding mechanism that is r -subset private, nearly maximally informative, and preserves signals for learning.

B. Related Work and Instance Encoding

Privacy preserving computation on encrypted data is a well-studied topic. When the data is distributed, techniques such as multiparty computing [6] and Lagrange coded computing [15] provide schemes with rigorous privacy guarantees that are often amenable to collusion of parties up to a certain threshold. Homomorphic encryption (HE) is a popular alternative for the two-party case, and several successful attempts of incorporating it into a learning framework have been reported [3, 8, 10]. Yet, these approaches are mired by the intricate nature of HE, as well as the drastic computation and communication overheads. In addition, HE provides computational rather than information-theoretic privacy.

Instance Encoding has been recently posed as an approach to circumvent the overhead of HE and multiparty computing while maintaining privacy (e.g., experimentally against known attacks), even though the terminology is yet somewhat unsettled. The award-winning work InstaHide [9] offered the first such discernible attempt. By linearly combining the private datapoints with public ones, along with random bit-flipping, [9] makes certain privacy claims and demonstrates marginal accuracy loss (although the former has been repeatedly challenged since [1, 4, 5]). Similar privacy claims have been made by NeuraCrypt [14], and then challenged by [2]. Additional data transformation mechanisms have been studied in [16, 17], analyzed via the PAC learning framework, and incorporated differential-privacy guarantees in the single data-owner case.

Prior works in this genre vary in their assumptions and problem setup. Variations include the existence of a public dataset that can be utilized for encoding [9], an encoding function that is available to the attacker [1], or an attacker which may freely choose the data distribution [1]. The authors are not aware of any prior attempts to formulate an instance encoding framework in an information-theoretic or algebraic way (see Remark 1) and the paper at hand does so based on the notions of perfect subset privacy and signal preservation.

We emphasize that our work addresses a two-party setting (data owner and a service provider), does not incorporate public datapoints, and relies on a choice of a random key

known only to the data owner. We use the term “instance encoding” to describe a signal-preserving privatization scheme whose output is a dataset of the same size as the original. As described in the next section, our goal is to guarantee perfect subset privacy and maximizing mutual information with the original dataset.

II. PRELIMINARIES AND PROBLEM SETUP

Let $X^n := (X_1, \dots, X_n)$ be a dataset of n samples, independently and identically distributed (i.i.d.) according to P_X . We assume that $\mathcal{X}$ is discrete, although the arguments can be readily adapted to the continuous case by quantization. The setup considers sharing a (possibly randomized) transformation of X^n to an external party subject to privacy and informativeness constraints. More specifically, the goal is to design a mechanism $P_{\tilde{X}^n|X^n}$ that generates $\tilde{X}^n$ from X^n , such that $\tilde{X}^n$ satisfies r -subset privacy (as defined next) while maintaining maximal $I(X^n; \tilde{X}^n)$.¹ To define the privacy notion, denote $[n] := \{1, \dots, n\}$ and let $\binom{[n]}{r}$ be the collection of all r -subsets of $[n]$. Also, for any $\mathcal{R} \subseteq \binom{[n]}{r}$, we use the shorthand $X_{\mathcal{R}} := (X_i)_{i \in \mathcal{R}}$.

Definition 1 (r -subset privacy). *Fix $r \in \mathbb{N}$. The privatization mechanism $P_{\tilde{X}^n|X^n}$ is said to satisfy r -subset privacy if*

$$I(X_{\mathcal{R}}; \tilde{X}^n) = 0, \quad \forall \mathcal{R} \in \binom{[n]}{r}, \quad (1)$$

where the underlying distribution is the appropriate marginal of $(X^n, \tilde{X}^n) \sim P_X^{\otimes n} P_{\tilde{X}^n|X^n}$.

Remark 1 (Differential privacy). *Differential privacy (DP) is a popular notion of statistical privacy. It guarantees that the change in the statistics of an output query Y (computed on the data) as a result of varying a single data entry, will be negligible. DP was recently shown to be (almost) equivalently expressible in information-theoretic terms as [7]²*

$$\sup_{i, P_{X^n}} I(Y; X_i | X_{[n] \setminus \{i\}}) \leq \epsilon.$$

The statistic that corresponds to our setting is $Y = \tilde{X}^n$.

Although r -subset privacy is reminiscent of the information-theoretic DP formulation (especially when $r = n - 1$), there are several notable differences. First, there is no distributional assumption under the DP framework (hence the supremum over P_{X^n}), while we assume that $X^n \sim P_X^{\otimes n}$ with P_X known. On the other hand, r -subset privacy is more flexible (and tunable) and guarantees zero mutual information, while DP trivializes when $\epsilon = 0$. Consequently, we see no natural and meaningful way to compare these two privacy paradigms.

III. GENERAL-PURPOSE PRIVATIZATION MECHANISM

We propose a general-purpose privatization mechanism; a learning-oriented adaptation of the presented scheme is

¹Currently, we do not specify or restrict the amount of external randomness the mechanism employs.

²More precisely, the mutual information constraint is implied by $(\epsilon, 0)$ -DP and implies $(0, \sqrt{2\epsilon})$ -DP.

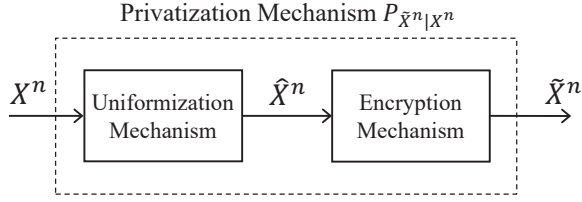


Fig. 1: End-to-end privatization scheme, as a composition of uniformization and encryption mechanisms.

addressed in Section IV. The scheme first uniformizes the data and then encrypts the uniform version, as shown in Fig. 1. Composing the uniformization and encryption mechanisms, informativeness and privacy are guaranteed end-to-end.

Throughout, we use $X, \hat{X}$, and $\tilde{X}$ for the original, uniformized, and encrypted data random variables, respectively. Lowercase letters, e.g., $x, \hat{x}$, and $\tilde{x}$, denote realizations from these variables, while calligraphic letters, e.g., $\mathcal{X}, \hat{\mathcal{X}}$, and $\tilde{\mathcal{X}}$ designate their alphabets. Similar notations are used for other random variables (e.g., for a random key K from an alphabet $\mathcal{K}$ and instances k).

A. Uniformization Mechanism

Consider the original data random variable $X \sim P_X$ (the dataset X^n consists of i.i.d. copies of X) and suppose it comprises m features, i.e., $X = (X(j))_{j=1}^m$. Partition these m features into ℓ mutually independent subsets (possibly with $\ell = 1$) that are collected into the random variables $S_1, \dots, S_\ell$. Accordingly, we represent $X = (S_1, \dots, S_\ell)$ and have $P_X = P_{S_1} \otimes \dots \otimes P_{S_\ell}$.

For a natural number p , we say that a discrete random variable is p -dyadic if the probability of every element in its support is an integer multiple of $\frac{1}{p}$.³ For example, a Bernoulli random variable with parameter $1/4$ is 4-dyadic, but not 2-dyadic. We assume that for each $i = 1, \dots, \ell$, the random variable S_i is p_i -dyadic for some p_i which is an integer power of two. With the assumption, we next show how to transform P_X to $\text{Unif}(\{0, 1\}^d)$ for some d , as defined shortly.

As each S_i is p_i -dyadic, for every $s \in \text{supp}(S_i)$ there exists an integer $n_{s,i}$ such that $\mathbb{P}(S_i = s) = n_{s,i}/p_i$. Partition $\{0, 1\}^{\log(p_i)}$ into $|\text{supp}(S_i)|$ disjoint subsets $\{\mathcal{T}_s^i\}_{s \in \text{supp}(S_i)}$ of respective sizes $\{n_{s,i}\}_{s \in \text{supp}(S_i)}$, which is possible since $\sum_{s \in \text{supp}(S_i)} n_{s,i} = p_i$. Set $d = \sum_{i=1}^{\ell} \log(p_i)$.

Given a datapoint $x = (s_1, \dots, s_\ell) \in \mathcal{X}$ (a realization of X), map it to a vector in $\{0, 1\}^d$ by mapping each s_i to a uniformly chosen element of $\mathcal{T}_{s_i}^i$ and concatenating the ℓ obtained vectors. Denote the resulting random variable by $\hat{X}$, and notice that it is a concatenation of ℓ mutually independent random variables, i.e., $\hat{X} = (u_1(S_1), \dots, u_\ell(S_\ell))$, where each $u_i(S_i)$ is given by mapping the respective s_i to an element of $\{0, 1\}^{\log(p_i)}$ as described above.

³Not to be confused with a dyadic random variable, in which all probabilities are of the form $1/2^i$ for some $i \in \mathbb{N}$.

Proposition 1 (Uniformization). *Let $X = (S_1, \dots, S_\ell)$ for mutually independent S_i 's. If each S_i is p_i -dyadic, where p_i is an integer power of two, then $\hat{X}$ is uniform over $\{0, 1\}^d$ with $d = \sum_{i=1}^{\ell} \log(p_i)$.*

Proof. Since the S_i 's are mutually independent, it suffices to prove that each $u_i(S_i)$ is uniform over $\{0, 1\}^{\log(p_i)}$. For $y \in \{0, 1\}^{\log(p_i)}$, let $\mathcal{T}_s^i$ be the subset in the partition of $\{0, 1\}^{\log(p_i)}$ to which it belongs, i.e., $y \in \mathcal{T}_s^i$. We thus have

$$\mathbb{P}(u_i(S_i) = y) = \frac{n_{s,i}}{p_i} \cdot \frac{1}{n_{s,i}} = \frac{1}{p_i},$$

because $\mathbb{P}(u_i(S_i) = y) = n_{s,i}/p_i$ by assumption and y is chosen uniformly from $\mathcal{T}_s^i$, a set of size $n_{s,i}$. $\square$

B. Encryption Mechanism

Let $\hat{X}^n := (\hat{X}_1, \dots, \hat{X}_n)$ denote the uniformized dataset, i.e., it is i.i.d. according to $P_{\hat{X}} = \text{Unif}(\{0, 1\}^d)$. Under this assumption, we view the datapoints as elements of $\mathbb{F}_{2^d}$, the finite field of size 2^d .

Fix $r \in [n]$, and let $K_1, \dots, K_r$ be independent and uniform random variables over $\mathbb{F}_{2^d}$. The key vector $K := (K_1, \dots, K_r)$ serves as external randomness in the encryption mechanism. For fixed elements $w_1, \dots, w_n \in \mathbb{F}_{2^d}$, define

$$\tilde{X}_i := \hat{X}_i + \sum_{j=1}^r w_i^{j-1} \cdot K_j, \quad i \in [n] \quad (2)$$

where '+' and ' $\cdot$ ' denote the respective operations in $\mathbb{F}_{2^d}$. In matrix form, (2) can be written as

$$(\tilde{X}_1, \dots, \tilde{X}_n) = (\hat{X}_1, \dots, \hat{X}_n, K_1, \dots, K_r) \cdot (I_n \ G^T)^T, \quad (3)$$

where I_n is the $n \times n$ identity over $\mathbb{F}_{2^d}$, and $G = G_{w^n}$ is an $r \times n$ Vandermonde matrix over $w_1, \dots, w_n$. The relation in (3) gives rise to a privatization mechanism, denoted by $P_{\tilde{X}^n|\hat{X}^n}$.

Theorem 1 below describes the privacy and informativeness properties of the mechanism $P_{\tilde{X}^n|\hat{X}^n}$. To state it, let $P_{\hat{X}|X}$ be the uniformization mechanism described in the previous section. The joint distribution of $(X^n, \hat{X}^n, \tilde{X}^n)$ is thus $P_X^{\otimes n} P_{\hat{X}|X}^{\otimes n} P_{\tilde{X}^n|\hat{X}^n}$; notice, in particular, that $(X^n, \hat{X}^n)$ are i.i.d. copies of $(X, \hat{X}) \sim P_X P_{\hat{X}|X}$. The overall privatization mechanism from the original dataset X^n to the privatized version $\tilde{X}^n$ is given by the marginal conditional distribution $P_{\tilde{X}^n|X^n}$ induced from the joint distribution above.

Theorem 1 (r -subset privacy and informativeness). *For any $r \in [n]$, if $\log n \leq d$ and $w_1, \dots, w_n \in \mathbb{F}_{2^d}$ are distinct, then the end-to-end privatization mechanism $P_{\tilde{X}^n|X^n}$:*

- (1) *satisfies r -point privacy, i.e., (1) holds; and*
- (2) *attains $I(X^n; \tilde{X}^n) \geq (n - r)H(X) - rH(\hat{X}|X)$.*

Proof outline. For (1), observe that $X_{\mathcal{R}} \rightarrow \hat{X}_{\mathcal{R}} \rightarrow \tilde{X}^n$ forms a Markov chain for any $\mathcal{R} \in \binom{[n]}{r}$. Hence it suffices to show that $\mathbb{P}(\hat{X}_{\mathcal{R}} = \hat{x}^r, \tilde{X}^n = \tilde{x}^n) = \mathbb{P}(\hat{X}_{\mathcal{R}} = \hat{x}^r) \mathbb{P}(\tilde{X}^n = \tilde{x}^n)$ for every $\hat{x}^r$ and $\tilde{x}^n$. This is proved by marginalizing on values of K , and observing that there exists a unique key k which decodes $(\tilde{x}_i)_{i \in \mathcal{R}}$ to $\hat{x}^r$. To prove (2), the chain rule

and (1) are used to show that $I(X^n; \tilde{X}^n) = (n-r)H(X) - H(X_{[n]\setminus\mathcal{R}}|\tilde{X}^n, X_{\mathcal{R}})$. The above fact about a unique key is used again to bound this expression from below. $\square$

Remark 2 (Theorem 1 assumptions). *When $r > 1$, the condition $\log n \leq d$ is necessary to guarantee that distinct $w_1, \dots, w_n$ exist (the condition can be removed for $r = 1$). One can treat the case of $r > 1$ and $\log n > d$ by padding the output of uniformization mechanism with $b > 0$ independent, uniform random bits, so that $\hat{X}$ is uniform over strings of length $d+b$, with $\log n \leq d+b$. This, however, results in larger $H(\hat{X}|X)$, and hence smaller $I(X^n; \tilde{X}^n)$, albeit b grows at most logarithmically with n .*

IV. LEARNING-ORIENTED PRIVATIZATION MECHANISM

Maintaining high mutual information between the original and private datasets is a holistic approach. It does not account for how that information could be extracted or using which processing algorithms. The instance encoding framework aims to address this issue and enable certain classes of algorithms to access the information through the privacy mechanism.

We next formulate a necessary condition for a learning algorithm to access that information. The goal is to capture the idea that learning tasks that were possible on the original dataset are still possible after it is privatized.

Definition 2 (Hypothesis class preservation). *Let $f : \mathcal{X}_1 \rightarrow \mathcal{X}_2$ be a mapping between feature spaces and consider two hypothesis classes $\mathcal{H}_1$ and $\mathcal{H}_2$ of functions from $\mathcal{X}_1$ or $\mathcal{X}_2$, respectively, to a label space $\mathcal{Y}$. The map f is said to be $(\mathcal{H}_1, \mathcal{H}_2)$ -preserving if for any $h_1 \in \mathcal{H}_1$ there exists $h_2 \in \mathcal{H}_2$ such that $h_1(x) = h_2 \circ f(x)$, for every $x \in \mathcal{X}_1$.*

Definition 2 concerns preserving learnability when mapping between two fixed hypothesis classes. However, different learning algorithms use different hypothesis classes (e.g., depending on architecture), and optimally, we would like to account for learnability of multiple models. To that end, we note that a variety of models can be represented as $h(x) = \tilde{h}(Wx^\top - v^\top)$, i.e., they first apply a linear transformation on the input and only then proceed with further processing. This includes, among others, linear classification, logistic regression, and feedforward neural networks. Restricting attention to such models, the next definition considers mappings that preserve the *signals* $wx^\top - b$, for an arbitrary weight vector w and scalar b , so as to enable learning the entire model.

Definition 3 (Signal preservation). *A map $f : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^{d_2}$ is signal preserving if for every $(w_1, b_1) \in \mathbb{R}^{d_1+1}$ there exists $(w_2, b_2) \in \mathbb{R}^{d_2+1}$, such that*

$$w_1 x^\top - b_1 = w_2 f(x)^\top - b_2, \quad \forall x \in \mathbb{R}^{d_1}.$$

If the privatization mechanism is signal-preserving, then any learning algorithm that is based on finding signals (say, to minimize a loss function), can be executed as-is on the privatized data. The optimization problem would now seek the new signals rather than the original ones (e.g., (w_2, b_2) instead of (w_1, b_1) in Def. 3). A potential disadvantage of this

approach is increased complexity and risk of overfitting, which are beyond the scope of this paper. One can readily verify that any invertible linear transform is signal preserving, and that signal-preserving maps are closed under composition.

We next devise a signal-preserving uniformization and encryption schemes. This results in an end-to-end signal preserving privatization mechanism due to the composition property.

A. Uniformization Mechanism

In the remainder of this paper we use $\{\pm 1\}$ to represent the Boolean field $\mathbb{F}_2$, instead of $\{0, 1\}$. For an integer p that is a power of two, let C_{p-1} be the *punctured Hadamard code* of length $p-1$; this is a set of p vectors in $\{\pm 1\}^{p-1}$, obtained by taking the rows of the $p \times p$ Sylvester-type Hadamard matrix, and omitting its all 1's column. That is, we define:

$$H_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_{2p} = \begin{pmatrix} H_p & H_p \\ H_p & -H_p \end{pmatrix}, \quad (4)$$

and for every p which is an integer power of two, we let $H_{p-1} \in \{\pm 1\}^{p \times (p-1)}$ be H_p with its all 1's column (leftmost) omitted. The set of rows of H_{p-1} is called a *punctured Hadamard code of order p* , and is denoted hereafter by $C_{p-1} \subseteq \{\pm 1\}^{p-1}$. It is well known that C_{p-1} is a Boolean subspace over $\mathbb{F}_2$ of dimension $\log(p)$.

As in Section III, we assume that the data distribution is decomposable as $X = (S_1, \dots, S_\ell)$, with the S_i 's mutually independent, and each S_i is p_i -dyadic for some integer power of two p_i . To obtain signal preservation, we uniformize each S_i on C_{p_i-1} (rather than on $\{0, 1\}^{\log(p_i)}$ as done in Section III).

Specifically, partition C_{p_i-1} to $|\text{supp}(S_i)|$ disjoint subsets $\{\mathcal{T}_s^i\}_{s \in \text{supp}(S_i)}$ of sizes $\{n_{s,i}\}_{s \in \text{supp}(S_i)}$, and let $d' = \sum_{i=1}^\ell (p_i - 1)$. A point $x = (s_1, \dots, s_\ell)$ drawn from P_X is mapped to an element of $\{\pm 1\}^{d'}$ by a process we call *random Hadamard encoding*, as follows: map each s_i to a uniformly random element in $\mathcal{T}_{s_i}^i$. Denote the resulting random variable by $\hat{X}_{\text{sp}}$, and let $\hat{X}_{\text{sp}}^n = (\hat{X}_{\text{sp},1}^n, \dots, \hat{X}_{\text{sp},n}^n)$ be the corresponding n -fold extension obtained by applying the above process to dataset X^n . The proof of the following proposition, which is omitted, follows the outline of the proof of Proposition 1, and then employs the invertibility over $\mathbb{R}$ of Hadamard matrices.

Proposition 2 (Signal-preserving uniformization). *Let $V := C_{p_1-1} \times \dots \times C_{p_\ell-1}$. Random Hadamard encoding is a signal-preserving uniformization scheme on V , i.e., $\hat{X}_{\text{sp}} \sim \text{Unif}(V)$.*

B. Encryption Mechanism

Simply replacing $\hat{X}_i$ by $\hat{X}_{\text{sp},i}$ in (2) does not guarantee r -point privacy, since $\hat{X}_{\text{sp},i}$ is not $\text{Unif}(\{\pm 1\}^{d'})$, but rather $\text{Unif}(V)$. Nevertheless, by properly representing $\mathbb{F}_{2^{d'}}$ as $\{\pm 1\}$ vectors, choosing the w_i 's in G_{w^n} judiciously, and drawing the random keys from $\text{Unif}(V)$ rather than $\text{Unif}(\{\pm 1\}^{d'})$, we obtain perfect r -point privacy.

Specifically, in what follows it is assumed that V represents a subspace of $\mathbb{F}_{2^{d'}}$ over a subfield $\mathbb{F}_{2^m}$ for some m such that $m|d'$ and $m \geq \log n$. That is, V is closed under addition and under multiplication by an element from $\mathbb{F}_{2^m}$. An explicit

choice of basis of $\mathbb{F}_{2^{d'}}$ by which V is indeed a subspace over $\mathbb{F}_{2^m}$ will be given in the full version of this paper. Define

$$\tilde{X}_{\text{sp},i} := \hat{X}_{\text{sp},i} + \sum_{j=1}^r w_i^{j-1} \cdot K_j, \quad i \in [n] \quad (5)$$

where ‘+’ and ‘·’ are the respective operations in $\mathbb{F}_{2^{d'}}$, $w_1, \dots, w_n \in \mathbb{F}_{2^m}$, and $K_j \sim \text{Unif}(V)$. The relation in (5) gives rise to a privatization mechanism $P_{\tilde{X}_{\text{sp}}^n | \hat{X}_{\text{sp}}^n}$. Together with the random Hamamrd encoding uniformization scheme $P_{\hat{X}_{\text{sp}}^n | X^n}$, we have $(X^n, \hat{X}_{\text{sp}}^n, \tilde{X}_{\text{sp}}^n) \sim P_X^{\otimes n} P_{\hat{X}_{\text{sp}}^n | X^n} P_{\tilde{X}_{\text{sp}}^n | \hat{X}_{\text{sp}}^n}$.

Theorem 2 (*r*-subset privacy and informativeness). *Fix $r \in [n]$ and let m satisfy $m | \gcd(d, d')$ and $\log n \leq m$. If $w_1, \dots, w_n \in \mathbb{F}_{2^m}$ are distinct and V is a subspace over $\mathbb{F}_{2^m}$, then the end-to-end privatization mechanism $P_{\tilde{X}_{\text{sp}}^n | X^n}^{\text{sp}}$:*

- 1) *satisfies r -point privacy: $I(X_{\mathcal{R}}; \tilde{X}_{\text{sp}}^n) = 0$, $\forall \mathcal{R} \in \binom{[n]}{r}$; and*
- 2) *attains $I(X^n; \tilde{X}_{\text{sp}}^n) \geq (n - r)H(X) - rH(\hat{X}_{\text{sp}} | X)$.*

Theorem 2 follows by an argument similar to that of Theorem 1, and is omitted due to space limitations.

Remark 3. *Similar to Remark 2, for $r = 1$ the relation in (5) does not require the existence of n distinct elements in $\mathbb{F}_{2^m}$. Thus, when $r = 1$ the above theorem holds even without the additional requirement on V , and without the existence of m .*

C. End-to-End Signal Preservation

By Proposition 2 and the composition property of signal-preserving maps, it suffices to show that the encryption mechanism $P_{\tilde{X}_{\text{sp}}^n | \hat{X}_{\text{sp}}^n}$ is signal-preserving to obtain end-to-end signal-preservation (i.e., for $P_{\tilde{X}_{\text{sp}}^n | X^n}$). Recall from Sections III-A and IV-A that $d = \sum_{i=1}^{\ell} \log(p_i)$, $d' = \sum_{i=1}^{\ell} (p_i - 1)$, and $P_X = P_{S_1} \otimes \dots \otimes P_{S_{\ell}}$ with each S_i as p_i -dyadic.

Proposition 3. *For $r = 1$, the encryption in (2) is signal-preserving. For $r > 1$ and $n \leq d$, the encryption in (2) is signal-preserving with probability at least $1 - (1/2 + o(1))^n$.*

Proof outline. For $r = 1$ the encryption mechanism amounts to multiplying by the matrix $\text{diag}(K_1)$ over $\mathbb{R}$, and the claim follows since $\text{diag}(K_1)$ is invertible. For $r > 1$, we have that both $\hat{X}_{\text{sp}}^n$ and $\tilde{X}_{\text{sp}}^n$ are uniform over V , which is a concatenation of punctured Hadamard codes. By employing the structure of Hadamard codes as Boolean subspaces, and using a recent result about the invertibility of random $\{\pm 1\}$ matrices [13], this fact is used to show that a sample of either $\hat{X}_{\text{sp}}^n$ or $\tilde{X}_{\text{sp}}^n$, seen as a matrix in $\mathbb{R}^{n \times d'}$, is full-rank with the aforementioned probability. Therefore, the encryption mechanism is invertible over $\mathbb{R}$, hence signal-preserving. $\square$

V. INFORMATIVENESS UPPER BOUND

An r -point privacy mechanism should keep $I(X^n; \tilde{X}^n)$ as large as possible. Since it is readily verified that any such mechanism has $I(X^n; \tilde{X}^n) \leq (n - r)H(X)$, we ask whether the additional uniformization penalties in Theorems 1-2 is necessary? To answer this question, we set up an information-theoretic framework that captures a broad class of r -point

privacy mechanisms and derive an upper bound on $I(X^n; \tilde{X}^n)$ under this framework. The set up is as follows.

- 1) **Dataset:** Recall that X^n contains i.i.d. copies of $X \sim P_X$.
- 2) **Random keys:** We assume the availability of individual and shared random keys. Let $B_1, \dots, B_n$ be individual keys (i.e., each B_i corresponds to X_i) that are conditionally i.i.d. given the datapoints according to $P_{B|X}$. Further let K be a shared key (i.e., the same K is used for all X_i 's) independent of (X^n, B^n) with distribution P_K .
- 3) **Encryption functions:** For every $i \in [n]$, let $f_i : \mathcal{X} \times \mathcal{B} \rightarrow \hat{\mathcal{X}}$ be a function that applies the individual key B_i to the point X_i , and set $\hat{X}_i = f_i(X_i, B_i)$. Similarly, let $g_i : \hat{\mathcal{X}} \times \mathcal{K} \rightarrow \mathcal{X}$ be a function that applies the joint key K to the output of f_i , and denote $\tilde{X}_i = g_i(\hat{X}_i, K)$.
- 4) **Reversibility:** We further require that schemes are reversible in the sense that given the keys $B_1, \dots, B_n, K$, one can retrieve X^n from its encrypted version $\tilde{X}^n$. Formally, we assume that $H(\hat{X}_i | \tilde{X}_i, K) = 0$ and $H(X_i | \hat{X}_i, B_i) = 0$ for all $i \in [n]$. Reversibility ensures that a trusted party (who has access to the key) can recover that data if they wish to do so.
- 5) **Subset privacy:** The mechanism satisfies the r -subset privacy requirement from (1).

Note that both mechanisms from Sections IV and III adhere to the above framework. We have the following upper bound on the informativeness of such mechanisms.

Theorem 3. *Any decodable scheme with $r = o(n/\log(n))$ and $|\mathcal{K}| \leq 2^{O(n^r)}$ satisfies*

$$I(X^n; \tilde{X}^n) \leq (n - r)H(X) - H(B_{\mathcal{R}} | X_{\mathcal{R}}, \tilde{X}_{\mathcal{R}}) + r \cdot \min_{x \in \mathcal{X}} H(B|x) - (n - r)H(X|\hat{X}) + o(1)$$

for every $\mathcal{R} \in \binom{[n]}{r}$.

Proof outline. Standard information-theoretic identities, similar to those in the proof of Theorem 1, are used to show that $I(X^n; \tilde{X}^n) = (n - r)H(X) - H(K | X_{\mathcal{R}}, \tilde{X}^n) + H(K | X^n, \tilde{X}^n) - (n - r)H(X|\hat{X})$. Then, using the decodability assumption we obtain $H(K | X_{\mathcal{R}}, \tilde{X}^n) = H(B_{\mathcal{R}} | X_{\mathcal{R}}, \tilde{X}_{\mathcal{R}})$. The argument relies on $\binom{n}{r}$ tri-partite graphs on $\mathcal{X}^r \times \hat{\mathcal{X}}^r \times \mathcal{X}^r$ induced by the functions f_i, g_i above; the labels of edges of this graph (keys B, K which map one vertex to another) are vertex disjoint. Lastly, one shows that $H(K | X^n, \tilde{X}^n) \leq r \cdot \min_{x \in \mathcal{X}} H(B|x) + o(1)$ by partitioning to typical and atypical datasets, with the latter contributing the $o(1)$ term. $\square$

In our schemes, outputs of the uniformization mechanism corresponding to different $x \in \mathcal{X}$ values are disjoint, and hence $H(X|\hat{X}) = 0$. Moreover, the terms $H(B_{\mathcal{R}} | X_{\mathcal{R}}, \tilde{X}_{\mathcal{R}})$ and $\min_{x \in \mathcal{X}} H(B|x)$ can be seen as quantifiers of the “uniformity” of X , as both can be nullified if X is uniform (e.g., by taking $B = 0$). As our schemes satisfy that $H(B_{\mathcal{R}} | X_{\mathcal{R}}, \tilde{X}_{\mathcal{R}}) = rH(B|X)$ and $H(X|\hat{X}) = 0$, they are off by an additive $r \cdot \min_{x \in \mathcal{X}} H(B|x)$ term from the above upper bound. In the full version of this paper, we present a (rather esoteric) variant that attains this bound with equality, and therefore, no better bound is possible in general.

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